

MATH1081 DISCRETE MATHEMATICS

Section 2

Integers, modular arithmetic
and relations

Number theory began as the study of important properties of the positive integers. Along with geometry, it is one of the oldest branches of mathematics. For many centuries number theory was studied principally for its own sake, and many of the greatest mathematicians have been fascinated by the extraordinary properties which lie concealed beneath the “obvious” surface of the integers.

In the eighteenth century, the domain of number theory began to expand to encompass various types of real and even complex numbers which are “integer-like” in some way, and with the transition from analogue to digital machinery in the late twentieth century number theory has become more and more important in technological applications. Number theory underlies most modern forms of data encryption; together with modern abstract algebra it provides methods for negating the effect of transmission errors when reading a disk, using an optical fibre or communicating through a wireless network.

So, every time you watch a DVD, send an email or tap a card to make a payment, you are relying on number theory to make it all work! It truly is an example of mathematics that you use every day.

The most obvious difference between the arithmetic of integers and that of real numbers is that whereas the quotient of two real numbers (with non-zero denominator) is always a real number, this need not be true of integers. For example, dividing 12 by 4 gives an integer result, but dividing 12 by 5 does not. The case when the quotient of two integers does give an integer is particularly important.

Definition. Let a and b be integers. If there is an integer m such that $b = am$ we say that a is a **factor** of b and write $a \mid b$.

Examples.

- $4 \mid 12$ since $12 = 4 \times 3$.
- $7 \mid 91$ since $91 = 7 \times 13$.
- $73 \mid 10001$ since $10001 = 73 \times 137$.
- The set of all (positive) factors of 12 is

$$\{ 1, 2, 3, 4, 6, 12 \} .$$

Note that factors are not affected by signs: for example, 4 is a factor of -12 and -4 is also a factor of 12 (and of -12). Much of the time we need only refer to positive factors.

Alternative terminology. If a is a factor of b we also say that

- b is a **multiple** of a ;
- a is a **divisor** of b ;
- a **divides** b ;
- b is **divisible** by a .

If a is not a factor of b we cross out the divisibility symbol thus:

$$a \nmid b .$$

For example, $234 \nmid 567$ because $567 = 234m$ is not true for any integer m . (Why?)

Caution! Do not confuse “ a divides b ” with “ a divided by b ”. The latter is a *number* and can be written as, for example,

$$\{ 12 \text{ divided by } 5 \} = 12 \div 5 = 12/5 = 2.4 .$$

However “ a divides b ” has no numerical value: it is just a *statement* which is either true or false. Do not mix up the *divisibility* symbol and the *division* symbol,

$$12 \mid 5 \quad \text{and} \quad 12/5 .$$

Comment. As you know, *division by zero* is meaningless. However *divisibility by zero* makes perfectly good sense, though it is of no great importance. **Exercise.** Why not?

Theorem. *Properties of divisibility.* Let a, b and c be integers.

- If $a \mid b$ and $a \mid c$ then $a \mid b + c$ and $a \mid b - c$.
- Let s, t be integers. If $a \mid b$ and $a \mid c$ then $a \mid sb + tc$.
- If $a \mid b$ and $b \mid c$ then $a \mid c$.

Proof. We shall prove the third part and leave the rest as exercises. Let a, b and c be integers such that $a \mid b$ and $b \mid c$. By definition,

$$b = am \quad \text{and} \quad c = bn$$

for some integers m and n . Combining these equations,

$$c = (am)n = a(mn) = ak ,$$

where $k = mn$. But m and n are integers, so k is an integer. Therefore $a \mid c$.

Exercises. Prove the first two parts of this theorem. You can also try questions 7 (a) and (b) from the problem book.

Note that you must always set out your proofs clearly and logically, and give reasons for everything you say. *Don't* just write down a string of equations without explanation: *a proof containing no words is normally a very poor proof!*

Divisibility tests. It is often useful to be able to tell quickly (without a calculator!) whether or not one number is a factor of another.

- Is 2 a factor of 438651972?
- Is 3 a factor of 438651972?
- Is 4 a factor of 438651972?
- Is 5 a factor of 438651972?
- Is 6 a factor of 438651972?
- Is 7 a factor of 438651972?
- Is 8 a factor of 438651972?
- Is 9 a factor of 438651972?
- Is 10 a factor of 438651972?
- Is 11 a factor of 438651972?
- Is 12 a factor of 438651972?
- Is 13 a factor of 438651972?
- Is 14 a factor of 438651972?

Definition. An integer n , greater than 1, is called a **prime** number if it has no (positive) factors other than n itself and 1. An integer greater than 1 which is not prime is called **composite**.

Examples.

- The first few primes are 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, ...
- There is no end to the list of primes; that is, the number of primes is infinite. This was known to Euclid (about 325 B.C.–265 B.C.), and will be proved in section 3 of this course.
- Primes of the form $2^p - 1$ are known as *Mersenne primes*. The largest number currently known to be prime is

$$2^{82589933} - 1 ,$$

discovered on 7 December 2018. It is a number of over 24 million digits and was discovered by means of the cooperative computing project GIMPS. For more information visit www.mersenne.org and primes.utm.edu.

- Primes of the form $2^n + 1$ are known as *Fermat primes* and have important connections with geometry. Only five Fermat primes are known: 3, 5, 17, 257 and 65537.
- Primes differing by 2 are called *twin primes*. Examples include

$$3, 5 ; \quad 5, 7 ; \quad 11, 13 ; \quad 17, 19$$

and

$$10000000000061, 10000000000063 .$$

It is suspected that there are infinitely many twin primes, but nobody knows for sure.

Sometimes it is convenient to express the concepts of prime and composite numbers in slightly different ways.

Lemma. Suppose that p is prime and $p = ab$. Then either $a = 1$ or $b = 1$.

Proof. Let p be prime. If $p = ab$ then $a \mid p$. But p , being prime, has no factors other than 1 and p . So either $a = 1$ or $a = p$, and in the latter case we have $b = 1$. Therefore, either $a = 1$ or $b = 1$.

Lemma. If an integer $n \geq 2$ is composite then it is a product of two integers greater than 1.

Proof. Suppose that n is composite. Then, by definition, n has a factor a such that $1 < a < n$. Since $a \mid n$ we have $n = ab$ for some integer b ; as $a < n$ we have $b > 1$. Thus n is the product of a and b , which are both integers greater than 1.

Comment. Note that we didn't say a composite number is a product of two *different* integers greater than 1. Have a think about $n = 25$.

Prime numbers are important because they are the basic “building blocks” of all the integers. If a positive integer is not itself prime it can be factorised into two smaller numbers. If these numbers are not prime they can be factorised further, and the process continues until all the factors are prime. For example,

$$60 = 6 \times 10 = 2 \times 3 \times 2 \times 5 .$$

What happens if we attempt to factorise 60 in a different way? We might try

$$60 = 15 \times 4 = 3 \times 5 \times 2 \times 2 .$$

Since the order of factors in a product is of no importance we shall say that these two factorisations are really the same. And if you start factorising 60 in any other way, you will always find that you get the same final result. **Exercise.** Try it! Begin with $60 = 2 \times 30$ or $60 = 3 \times 20$.

This property holds for all positive integers, a fact whose importance is acknowledged by a suitable name.

Theorem. *The Fundamental Theorem of Arithmetic.* Any positive integer n can be factorised as a product of primes. Moreover, a given n has only one such factorisation, as long as the order of the prime factors is regarded as irrelevant.

Proof. To show that there is a prime factorisation we essentially follow the above example with $n = 60$. A formal proof would use mathematical induction: you will see this done in section 3 of the course, or you could try it now yourself if you learned induction at school. To show that there is *only one* factorisation is harder and probably beyond the scope of MATH1081.

Comments.

- We shall sometimes “standardise” the prime factorisation of an integer by writing the factors in increasing order and collecting like terms into powers, for example,

$$60 = 2 \times 2 \times 3 \times 5 = 2^2 \times 3 \times 5 .$$

However this is by no means essential.

- If n is prime we regard it as a “product” of just one prime (itself).
- By convention, $n = 1$ is regarded as a “product” of *no* primes!
- The Fundamental Theorem answers the commonly asked question “Why don’t we call 1 a prime?”

Examples. Some more prime factorisations:

$$1000 = 2^3 \times 5^3 ; \quad 1001 = 7 \times 11 \times 13 ; \quad 1002 = 2 \times 3 \times 167 .$$

Exercise. Find the prime factorisation of 345, and of 567.

Testing for primes. Let's work out whether or not 101 is prime. The obvious way to do this is to see whether or not it is a multiple of any of the numbers $2, 3, 4, 5, 6, \dots, 100$: if none of these is a factor then, by definition, 101 is prime, while if one of them is a factor then it is composite. However, this means we have to carry out 99 calculations. Can we do any better? Let's take it one step at a time.

- 101 is not a multiple of 2 because...
- 101 is not a multiple of 3 because...
- There is no need to test 4: if 101 were to be a multiple of 4 it would have to be a multiple of 2, and we have already checked that this is not the case. For the same reason there is no need to check 6, 8, 10 and so on.
- 101 is not a multiple of 5 because...
- 101 is not a multiple of 7 – check this!
- There is no need to test 9 because we have already ruled out 3. Similarly, there is no need to test 12, 15, 18,
- The next number to test is 11. **But** if 101 is composite then it is a product of two integers, $101 = ab$, where $a > 1$ and $b > 1$. Moreover, we know that $a \geq 11$ and $b \geq 11$, because we have already ruled out all smaller factors. Hence $ab \geq 11 \times 11$, that is, $101 \geq 121$, which is impossible.
- Therefore, 101 is prime.

The ideas of the previous example can be used to prove the following result, which is important for testing whether or not a number is prime.

Theorem. Let n be a positive integer. If n is composite, then n has a factor c such that $1 < c \leq \sqrt{n}$.

Proof. Let n be composite; then we can write $n = ab$, where $1 < a < n$. If $a \leq \sqrt{n}$ we can take $c = a$. If not then

$$n > a > \sqrt{n} ;$$

since $b = n/a$ it follows that

$$1 < b < \sqrt{n} ,$$

and we can take $c = b$.

Corollary. Any composite number n has a prime factor less than or equal to $\sqrt{n}$.

Proof. Let n be composite; let c be a factor of n with $1 < c \leq \sqrt{n}$; let p be a prime factor of c . Then p is a factor of n and $p \leq c \leq \sqrt{n}$.

Corollary. *Prime testing by trial division.* Let n be a positive integer greater than 1. If n has no prime factor less than or equal to $\sqrt{n}$, then n is prime.

Comment. You will learn more about *proof by division into cases* and about using the *contrapositive* in section 3 of this course.

Exercises. Is 161 prime? Is 163 prime? How about 167 and 169?

Definition. Let a and b be integers, not both of which are zero. Any integer d such that $d \mid a$ and $d \mid b$ is called a **common divisor** or a **common factor** of a and b . The largest such d is called the **greatest common divisor** of a and b .

If the greatest common divisor of a and b is 1 we say that a and b are **coprime** or **relatively prime** to each other.

Let a and b be positive integers. Any positive integer m such that $a \mid m$ and $b \mid m$ is called a **common multiple** of a and b . The smallest such m is called the **least common multiple** of a and b .

We write $\gcd(a, b)$ and $\text{lcm}(a, b)$ for the greatest common divisor and least common multiple of a and b .

Examples.

- The factors of 12 and of 42 are

$$\{1, 2, 3, 4, 6, 12\} \quad \text{and} \quad \{1, 2, 3, 6, 7, 14, 21, 42\}$$

respectively. Therefore their common divisors are 1, 2, 3 and 6 and we have

$$\gcd(12, 42) = 6 .$$

The multiples of these two numbers are

$$\{12, 24, 36, 48, 60, 72, 84, \dots\} \quad \text{and} \quad \{42, 84, \dots\} ,$$

and therefore

$$\text{lcm}(12, 42) = 84 .$$

- It is fairly clear that listing all the factors of two numbers would become very time-consuming if they were larger. We can cut down the work by writing numbers in terms of their prime factorisations. For example,

$$14175 = 3^4 \times 5^2 \times 7 \quad \text{and} \quad 16758 = 2 \times 3^2 \times 7^2 \times 19 .$$

To find the greatest common divisor of these two numbers we multiply all the prime factors common to both:

$$\gcd(14175, 16758) = 3^2 \times 7 = 63 .$$

To find the least common multiple we need the smallest product of prime factors which includes everything in both numbers:

$$\begin{aligned} \text{lcm}(14175, 16758) &= 2 \times 3^4 \times 5^2 \times 7^2 \times 19 \\ &= 3770550 . \end{aligned}$$

- If b is positive and is a factor of a , what is $\gcd(a, b)$?
- What is $\gcd(0, b)$ if $b > 0$?
- And what happens if we try to compute $\gcd(0, 0)$?
- What happens if we attempt to find $\text{lcm}(a, b)$ with either a or b equal to zero, for example, $\text{lcm}(0, 5)$?
- If a and b are positive integers then

$$\gcd(a, b) \text{lcm}(a, b) = ab .$$

We have seen how to test whether a number is prime or not by trial divisions, and how to compute the greatest common divisor of two numbers by using their prime factorisations. Let's try this for

$$m = 2^{678} + 3^{789} \quad \text{and} \quad n = 2^{345} + 3^{456} ,$$

numbers of 376 digits and 218 digits respectively. A bit big to do in my head, so I asked **Maple** instead.

I typed `ifactor(m)` and let **Maple** run for 100 seconds – no result. The same happened for n even though it's (a bit) smaller. So I tried to find out if m is prime using the command `isprime(m)`, and **Maple** instantaneously returned **false**; using a similar instruction I found that n is not prime either. Then I asked for the greatest common divisor directly by typing `igcd(m,n)` and immediately got the answer 1.

How is this possible?? To show that m is prime by trial divisions we have to check every possible prime divisor up to $\sqrt{m}$; to factorise a number we shall usually need fewer trials, because once we have found one factor it reduces the number that remains to be tested. So it would appear that factorisation should be easier than primality testing. Finding a greatest common divisor requires two factorisations: surely this must be twice as hard as finding one factorisation?

The answer to the above puzzles is that primality testing and calculation of greatest common divisors do not have to be done by trial divisions. There are superior methods which make these computations much faster. We shall look briefly at the problem of primality testing later (for more details you should take MATH2400), and shall see how to find greatest common divisors by **Euclid's algorithm**.

Playtime.

- Log into **Maple**, enter some large “random” numbers (say a couple of hundred digits) and see if you can factorise them completely in a reasonable time. If not, see how long it takes to find their greatest common divisor.
- If you type `factor(m)` instead of `ifactor(m)` you get an answer straight away! Unfortunately it is wrong. Can you explain why?

Early in school you probably learned integer division, with an integer divided by an integer giving an integer quotient and remainder. Later you may have been told that this is not “proper” division and that you should calculate a fractional or decimal quotient instead of the remainder. In fact, dividing integers to get a quotient and remainder is much more important in number theory than “proper” division.

Theorem. *The Division Algorithm.* Let a be an integer and b a positive integer. Then there is a unique pair of integers q, r such that

$$a = qb + r \quad \text{and} \quad 0 \leq r < b .$$

Proof. Let a and b be positive integers; take $q = \lfloor a/b \rfloor$ and $r = a - bq$. Then clearly $a = bq + r$; also

$$q \leq \frac{a}{b} < q + 1 \quad \Rightarrow \quad 0 \leq \frac{a}{b} - q < 1 \quad \Rightarrow \quad 0 \leq r < b .$$

We’ll defer the proof that there is only one suitable pair of values for q, r until Section 3 of the course.

Comment. Sometimes the remainder when a is divided by b is written as

$$r = a \bmod b .$$

There is a short question on this notation in Problem Set 2, but I will generally **not** be using it in lectures.

Examples.

- Find the quotient and remainder when 1001 is divided by 101. *Answer.* We have $1001 = 9 \times 101 + 92$, so the quotient is 9 and the remainder is 92.
- What are the quotient and remainder when 101 is divided by 1001?
- Find the quotient and remainder when 4321 is divided by 1234.
- Find the remainder when 438651972 is divided by 7.
- A bit of care is required when a is negative. If we divide -123 by 12, the quotient is *not* -10 and the remainder is *not* -3 .
- It is not hard to see that b is a factor of a if and only if dividing a by b gives zero remainder.

Comment. There is nothing especially wrong with dividing to give a *negative* remainder, for example,

$$1001 = 10 \times 101 - 9 ;$$

indeed, this can sometimes simplify calculations. However, for consistency we shall generally stick with non-negative remainders. You can prove the following for yourself.

Theorem. *Division with remainder of least absolute value.*
Let a be an integer and b a positive integer. Then there exists a unique pair of integers q, r such that

$$a = qb + r \quad \text{and} \quad -\frac{b}{2} < r \leq \frac{b}{2} .$$

There is an important relation between the Division Algorithm and greatest common divisors.

Theorem. Let a , b , q and r be integers such that $a = qb + r$. Then

$$\gcd(a, b) = \gcd(b, r) .$$

Proof. Let S be the set of common divisors of a and b , and T the set of common divisors of b and r . We shall prove that $S = T$.

So, let $d \in S$. Then $d \mid a$ and $d \mid b$, so $d \mid a - qb$ (using the second part of the theorem on page 5). That is, $d \mid r$. Since $d \mid b$ and $d \mid r$ we have $d \in T$; this proves that $S \subseteq T$.

The proof that $T \subseteq S$ is very similar; hence, $S = T$. Since a, b and b, r have exactly the same common divisors, they certainly have the same *greatest* common divisor.

Comment. Remember that usually the easiest way to prove that two sets are equal is to show that each is a subset of the other.

The importance of this result is that if we divide a by b in the usual way, the remainder r will be less than b . Therefore we can replace the problem of finding the greatest common divisor of two given numbers by that of finding the greatest common divisor of two smaller numbers, which, naturally, will be easier. Applying the Division Algorithm repeatedly, we may calculate the greatest common divisor of 16758 and 14175 thus:

$$16758 = 1 \times 14175 + 2583$$

$$\text{so } \gcd(16758, 14175) = \gcd(14175, 2583);$$

$$14175 = 5 \times 2583 + 1260$$

$$\text{so } \gcd(14175, 2583) = \gcd(2583, 1260);$$

$$2583 = 2 \times 1260 + 63$$

$$\text{so } \gcd(2583, 1260) = \gcd(1260, 63);$$

$$1260 = 20 \times 63 .$$

Now the last equation shows that 63 is a factor of 1260; putting all our information together shows that

$$\gcd(16758, 14175) = \gcd(1260, 63) = 63 ,$$

as we calculated earlier by using prime factorisations. And then by using a result from page 14 we have

$$\text{lcm}(16758, 14175) = \frac{16758 \times 14175}{\gcd(16758, 14175)} = 3770550 .$$

The above procedure is known as the **Euclidean Algorithm**.

Theorem. *The Euclidean Algorithm.* Let a and b be positive integers; suppose that

$$a = q_1b + r_1$$

$$b = q_2r_1 + r_2$$

$$r_1 = q_3r_2 + r_3$$

$$\vdots$$

$$r_{n-2} = q_nr_{n-1} + r_n$$

$$r_{n-1} = q_{n+1}r_n ,$$

where $q_1, q_2, \dots, q_{n+1}$ and $r_1, r_2, \dots, r_n$ are positive integers. Then

$$\gcd(a, b) = r_n .$$

Proof. Using the Division Algorithm repeatedly we have

$$\gcd(a, b) = \gcd(b, r_1) = \gcd(r_1, r_2) = \dots = \gcd(r_{n-1}, r_n) .$$

But the last line of the Euclidean Algorithm shows that $r_n \mid r_{n-1}$; thus $\gcd(r_{n-1}, r_n) = r_n$, and this completes the proof.

Example. Use the Euclidean Algorithm to find the greatest common divisor of 854 and 651.

Solution. We have

$$854 = 1 \times 651 + 203$$

$$651 = 3 \times 203 + 42$$

$$203 = 4 \times 42 + 35$$

$$42 = 1 \times 35 + 7$$

$$35 = 5 \times 7 ,$$

so $\gcd(854, 651) = 7$.

Exercises. Use the Euclidean Algorithm to prove that 1001 and 101 are relatively prime, and to find the greatest common divisor of 735 and 539.

Solutions.

Problem. In Martian Rules football the score consists of goals and behinds, each worth a certain number of points. In last season's MFL Grand Final one of the sides scored 79 goals and 45 behinds for a total of 5347 points. (MFL matches take a *looooong* time!) If a goal is worth more than a behind, how many points are scored for each?

Solution. Suppose that a goal is worth x points and a behind y points. Then

$$79x + 45y = 5347 .$$

Now if x and y are real numbers this equation is very easy to solve and there will be an infinite number of solutions. However, it seems reasonable to assume that x and y are positive integers (though you never know with Martians), and this makes things a bit harder. We'll look at some simple cases first and return to the MFL problem later.

So, first let's try to solve the equation

$$854x + 651y = 7 ,$$

where x and y are integers. This might look just as hard as the previous one, but in fact we can find a solution almost immediately as a by-product of the Euclidean Algorithm on page 21! Using the equations from the bottom of the page and rewriting to make the remainder the subject in each case,

$$\begin{aligned} 7 &= 42 - 1 \times 35 \\ 35 &= 203 - 4 \times 42 \\ 42 &= 651 - 3 \times 203 \\ 203 &= 854 - 1 \times 651 ; \end{aligned}$$

now substituting each of these equations into the previous one, expanding and collecting terms at each step,

$$\begin{aligned} 7 &= 42 - (203 - 4 \times 42) \\ &= -203 + 5 \times 42 \\ &= -203 + 5(651 - 3 \times 203) \\ &= 5 \times 651 - 16 \times 203 \\ &= 5 \times 651 - 16(854 - 651) \\ &= -16 \times 854 + 21 \times 651 . \end{aligned}$$

Notice that we have just calculated

$$-16 \times 854 + 21 \times 651 = 7 ,$$

and we want

$$854x + 651y = 7 ;$$

comparing these equations shows that a possible solution is

$$x = -16 \quad \text{and} \quad y = 21 .$$

Having solved the previous example, we should find it easy to find solutions of the following equations:

- $854x + 651y = 70$ has a solution $x =$ and $y =$;
- $854x + 651y = 700$ has a solution $x =$ and $y =$;
- $854x + 651y = 35$ has a solution $x =$ and $y =$;
- $854x + 651y = -14$ has a solution $x =$ and $y =$.

How about

$$854x + 651y = 53 \text{ ?}$$

This equation has no solutions in which x and y are integers. Remember that 854 and 651 have greatest common divisor 7; if there were a solution of the equation then we would have $7 \mid \text{LHS}$ and therefore $7 \mid \text{RHS}$. But 7 is not a factor of 53 and so this is impossible.

Theorem. *The Bézout property.* Consider the equation

$$ax + by = c , \tag{*}$$

where a, b and c are integers, with a and b not both zero. Write d for the greatest common divisor of a and b .

- If $c = d$ then $(*)$ has a solution in integers x, y .
- If c is a multiple of d then $(*)$ has a solution in integers.
- If c is not a multiple of d then $(*)$ has no solution in integers.

Proof. The proof of the first part is moderately hard – see question 11 in the problem book, which guides you through it with some hints. Assuming that this is known, we prove the remaining parts.

If c is a multiple of d we can write $c = dm$ for some integer m . Now from the first part of this theorem we know that $ax + by = d$ has a solution, say

$$ax_0 + by_0 = d .$$

Therefore

$$a(mx_0) + b(my_0) = md = c ,$$

and so $(*)$ has a solution $x = mx_0, y = my_0$.

Suppose that c is not a multiple of d . If x and y are integers then $d \mid a, d \mid b$ and hence $d \mid ax + by$. In $(*)$, therefore, $d \mid \text{LHS}$ while $d \nmid \text{RHS}$, and so there is no solution.

Exercises. Find a solution in integers of each of the following equations, *or* explain why they have no solution.

- $30x + 73y = 1$
- $91x + 49y = 2$
- $42x - 99y = 6$
- $12345678x + 98765432y = 123456789$

Solution.

Given an equation $ax + by = c$, we have seen how to find a solution in integers (if there is one). However, if an equation of this form has a solution, it will actually have many solutions (compare simultaneous equations in MATH1131), and we can write them all down in terms of a parameter.

Theorem. *Solving linear equations in integers.* Let a, b and c be integers with a and b relatively prime; suppose that the equation

$$ax + by = c \tag{*}$$

has a solution $x = x_0, y = y_0$. Then all solutions of this equation are given by the formulae

$$x = x_0 + \lambda b, \quad y = y_0 - \lambda a \quad \text{where} \quad \lambda \in \mathbb{Z}.$$

Proof. Suppose that x, y form a solution of (*). Subtracting the equations

$$ax + by = c \quad \text{and} \quad ax_0 + by_0 = c$$

gives $ax - ax_0 + by - by_0 = 0$, which can be written as

$$a(x - x_0) = -b(y - y_0). \tag{**}$$

Thus b is a factor of $a(x - x_0)$; but a and b are relatively prime and so b is a factor of $x - x_0$. (You are asked to prove this fact in problem 12.) So we have $x - x_0 = \lambda b$ for some integer λ ; substituting back into (**) yields

$$x = x_0 + \lambda b \quad \text{and} \quad y = y_0 - \lambda a$$

as claimed. Conversely, basic algebra shows that if these conditions hold then $ax + by = c$.

Comments.

- In fact, we can write these solutions in vector form

$$x = \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \end{pmatrix} + \lambda \begin{pmatrix} b \\ -a \end{pmatrix} \quad \text{where } \lambda \in \mathbb{Z} ,$$

which makes it look even more like linear algebra!

- Employing the Euclidean Algorithm to find a solution of the Martian Rules Football problem, and using the above theorem to write down all solutions, we have

$$x = 21388 + 45\lambda , \quad y = -37429 - 79\lambda$$

for some integer λ . Substituting these expressions into the inequalities $x > y > 0$ and solving gives $\lambda = -474$; hence

$$x = 58 \quad \text{and} \quad y = 17 .$$

Thus a goal is worth 58 points and a behind 17 points.

Exercise. Fill in the details of the working for this problem!

- If a and b have greatest common factor $d > 1$, and $ax + by = c$ has a solution, then d must be a factor of c . After dividing both sides by d we can apply the above theorem. For example, calculations from pages 21, 24 and 25 show that $\gcd(854, 651) = 7$ and that

$$854x + 651y = 35 \tag{*}$$

has a solution $x_0 = -80, y_0 = 105$. Dividing both sides of (*) by 7 gives

$$122x + 93y = 5 ;$$

since 122 and 93 are coprime the above theorem gives

$$x = -80 + 93\lambda , \quad y = 105 - 122\lambda \quad \text{where } \lambda \in \mathbb{Z} .$$

There is an important alternative way of thinking about divisibility relations.

Definition. Let m be an integer. Two integers a and b are said to be **congruent modulo m** , written

$$a \equiv b \pmod{m} ,$$

if $m \mid a - b$.

Examples.

- $12 \equiv 2 \pmod{10}$, $123 \equiv 3 \pmod{10}$ and $12345 \equiv 98765 \pmod{10}$.
- Find a single digit number d such that

$$1234567 \times 2345678 + 3456789 \equiv d \pmod{10} .$$

- $21 \equiv 0 \pmod{7}$ and $21 \equiv -21 \pmod{7}$. However $20 \not\equiv -20 \pmod{7}$.

Comment. For any a, b and m we have

$$m \mid a - b \quad \text{if and only if} \quad -m \mid a - b ,$$

that is,

$$a \equiv b \pmod{m} \quad \text{if and only if} \quad a \equiv b \pmod{-m} .$$

So considering negative moduli gives us nothing extra and we may as well assume that $m \geq 0$. In fact the cases $m = 0$ and $m = 1$ are of little interest (why?) and we shall generally take $m \geq 2$.

There are various different ways of expressing the idea of congruence. Let a, b and m be integers. Then

- $a \equiv b \pmod{m}$;
- $m \mid a - b$;
- $a = b + km$ for some integer k ;
- a and b have the same remainder when divided by m

all mean the same thing.

Proof. The first two are equivalent by definition; the third is not hard and is left as an exercise. To show that the fourth statement is equivalent to the others, first assume that a and b have the same remainder when divided by m . Then we can write

$$a = q_1m + r \quad \text{and} \quad b = q_2m + r$$

for some q_1, q_2 and r ; thus

$$a - b = q_1m - q_2m = (q_1 - q_2)m ,$$

which is divisible by m . Conversely, suppose that $m \mid a - b$ and that a divided by m leaves remainder r . Then

$$a - b = km \quad \text{and} \quad a = qm + r$$

for some integers k and q ; therefore

$$b = a - km = qm + r - km = (q - k)m + r$$

and so the remainder when b is divided by m is also r .

Comment. You will learn more about “if and only if” proofs in section 3 of the course.

What we have just done is, of course, really nothing new: a congruence is just another way of expressing divisibility. However, congruence notation is very convenient because it looks very much like equality, and handling congruences is indeed similar in many ways to handling equations.

Theorem. *Properties of congruences.*

- Let a, b, c, d, m be integers. If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$ then
 - $a + c \equiv b + d \pmod{m}$;
 - $a - c \equiv b - d \pmod{m}$;
 - $ac \equiv bd \pmod{m}$.
- If $a \equiv b \pmod{m}$ and $c \in \mathbb{Z}$ then $ca \equiv cb \pmod{m}$.
- If $a \equiv b \pmod{m}$ and $n \geq 0$ then $a^n \equiv b^n \pmod{m}$.
- If $a \equiv b \pmod{m}$ and $n \mid m$ then $a \equiv b \pmod{n}$.

Proof. We prove the third and last of these six properties; please try the other four as exercises. So, let a, b, c, d, m be integers such that $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$. Then

$$a = b + k_1m \quad \text{and} \quad c = d + k_2m$$

for some integers k_1, k_2 . Multiplying and rearranging,

$$\begin{aligned} ac &= bd + k_1md + bk_2m + k_1k_2m^2 \\ &= bd + (k_1d + bk_2 + k_1k_2m)m ; \end{aligned}$$

but $k_1d + bk_2 + k_1k_2m$ is an integer, and so $ac \equiv bd \pmod{m}$.

To prove the final claim, suppose that $a \equiv b \pmod{m}$ and $n \mid m$. Then $n \mid m$ and $m \mid a - b$, so $n \mid a - b$ and hence $a \equiv b \pmod{n}$.

Problem. Here is a really *really* easy question: what are the last two digits of 1234567?

Problem. ...and here is a harder one: what are the last two digits of $7^{1234567}$?

This type of problem is closely connected with modular arithmetic. In fact, we can say that the last two digits of 1234567 are 67 because

$$1234567 \equiv 67 \pmod{100} .$$

In other words, the question “what are the last two digits of...” is essentially the same as “simplify...modulo 100.”

It still seems like a lot of work to work out $7^{1234567}$. But if we start off by trial and error, calculating powers of 7 and simplifying modulo 100, we soon discover a useful fact:

$$7^1 \equiv 7 , \quad 7^2 \equiv 49 , \quad 7^3 \equiv 343 \equiv 43 , \quad 7^4 \equiv 301 \equiv 1 .$$

It turns out that $7^4 \equiv 1 \pmod{100}$. Using one of the “equality-like” properties on page 31, it is now easy to see that, for example,

$$7^8 = (7^4)^2 \equiv 1^2 \equiv 1 \pmod{100}$$

and

$$7^{444} = (7^4)^{111} \equiv 1^{111} \equiv 1 \pmod{100} .$$

So to answer the second problem above we only need to carry out an easy division and then do some simple algebra:

$$\begin{aligned} 7^{1234567} &= 7^{4 \times 308641 + 3} = (7^4)^{308641} \times 7^3 \\ &\equiv 1^{308641} \times 43 \equiv 43 \pmod{100} . \end{aligned}$$

The last two digits of $7^{1234567}$ are 43.

Exercise. Simplify $10^{123456789}$ modulo 41.

Solution.

Further examples. Simplifying a^n modulo m becomes quite easy if there is a small number k such that $a^k \equiv 1 \pmod{m}$. There are other cases which are similar but not identical.

- Simplify 5^{115511} modulo 29. Here we calculate

$$5 \equiv 5, \quad 5^2 \equiv 25 \equiv -4, \quad 5^3 \equiv -20 \equiv 9, \quad 5^4 \equiv 45 \equiv 16, \\ 5^5 \equiv 80 \equiv -7, \quad 5^6 \equiv -35 \equiv -6, \quad 5^7 \equiv -30 \equiv -1$$

modulo 29. Therefore

$$5^{115511} = 5^{7 \times 16501 + 4} = (5^7)^{16501} \times 5^4$$

and hence

$$5^{115511} \equiv (-1)^{16501} \times 16 \equiv -16 \equiv 13 \pmod{29}.$$

- Find the last two digits of 6^{54321} . Following the ideas of previous examples, we want a positive integer n such that

$$6^n \equiv 1 \pmod{100} \quad \text{or} \quad 6^n \equiv -1 \pmod{100}.$$

However, there is no such n ! (Why?) Nevertheless, the first few powers of 6 modulo 100 are

$$6^2 \equiv 36, \quad 6^3 \equiv 16, \quad 6^4 \equiv 96, \\ 6^5 \equiv 76, \quad 6^6 \equiv 56, \quad 6^7 \equiv 36.$$

We notice that 6^7 is the same as 6^2 , and so the remainders modulo 100 will repeat every five steps from here on:

$$6^8 \equiv 6^3 \equiv 16, \quad 6^9 \equiv 6^4 \equiv 96, \dots$$

Since $54321 \equiv 1 \equiv 6 \pmod{5}$ we have

$$6^{54321} \equiv 6^{54316} \equiv 6^{54311} \equiv \dots \equiv 6^6 \pmod{100},$$

and so the last two digits of 6^{54321} are 56.

Note that $6^6 \not\equiv 6^1 \pmod{100}$: the above pattern *does not* hold for smaller powers.

Solving congruences. In many problems of arithmetic, one or more of the quantities involved is initially unknown. Typically, we set up a system of equations in the unknowns and solve them using familiar algebraic techniques.

The same is true regarding congruences. For example, we might wish to find all x satisfying the condition

$$79x \equiv 5347 \pmod{45} .$$

Now by definition of a congruence, this is the same as saying that $79x = 5347 + 45y$ for some integer y ; in other words,

$$79x - 45y = 5347 .$$

But this is (almost) identical with the Martian Rules Football problem from page 23! Recalling the solution found on page 29 we have $x = 21388 + 45\lambda$ for any integer λ ; and this can be rewritten as a congruence

$$x \equiv 21388 \pmod{45} .$$

Finally, we can simplify this to give

$$x \equiv 13 \pmod{45} .$$

Further examples.**Example 1.** We solve the congruence

$$23x \equiv 1 \pmod{30} ,$$

showing all necessary working. The congruence is equivalent to the equation $23x = 1 + 30y$, that is,

$$23x - 30y = 1 \tag{*}$$

where y is an integer. Applying the Euclidean algorithm to 30 and 23 we have

$$30 = 1 \times 23 + 7$$

$$23 = 3 \times 7 + 2$$

$$7 = 3 \times 2 + 1 ,$$

and reversing this to obtain a Bézout identity gives

$$\begin{aligned} 1 &= 7 - 3 \times 2 \\ &= 7 - 3 \times (23 - 3 \times 7) \\ &= -3 \times 23 + 10 \times 7 \\ &= -3 \times 23 + 10(30 - 23) \\ &= 10 \times 30 - 13 \times 23 . \end{aligned}$$

Since $\gcd(23, 30) = 1$, the general solution of (*) has $x = -13 - 30\lambda$ with $\lambda \in \mathbb{Z}$ (and we can find y too, but it's not important). This equation is equivalent to the congruence

$$x \equiv -13 \pmod{30} ,$$

or

$$x \equiv 17 \pmod{30} ,$$

which is the solution of the given congruence.

Example 2. To solve the congruence

$$23x \equiv 11 \pmod{30}$$

we rewrite it as the equation

$$23x - 30y = 11, \quad y \in \mathbb{Z},$$

and follow through exactly the same working as in the previous example to obtain

$$1 = 10 \times 30 - 13 \times 23.$$

Multiplying both sides by 11 gives

$$11 = 110 \times 30 - 143 \times 23,$$

and (recalling once again that 23 and 30 are coprime) we have the solution $x = -143 - 30\lambda$ where $\lambda \in \mathbb{Z}$. Therefore the solution of the congruence is

$$x \equiv -143 \pmod{30},$$

which can (and should) be simplified to

$$x \equiv 7 \pmod{30}.$$

Example 3. Solve

$$4433x \equiv 44332211 \pmod{30}.$$

This can be solved by using exactly the same method as above, but it would be more sensible to simplify the congruence first.

Exercise. How?

Example 4. Solve

$$182x \equiv 57 \pmod{403} .$$

We need

$$182x - 403y = 57 ; \tag{*}$$

the Euclidean algorithm gives

$$403 = 2 \times 182 + 39$$

$$182 = 4 \times 39 + 26$$

$$39 = 1 \times 26 + 13$$

$$26 = 2 \times 13 .$$

Hence $\gcd(182, 403) = 13$. However $13 \nmid 57$, so $(*)$ has no solution and the congruence $182x \equiv 57 \pmod{403}$ has no solution.

Example 5. Solve

$$52x \equiv 4 \pmod{60} .$$

Here we are looking at

$$52x - 60y = 4 , \tag{*}$$

and the appropriate Euclidean algorithm is

$$60 = 1 \times 52 + 8$$

$$52 = 6 \times 8 + 4$$

$$8 = 2 \times 4 .$$

Since 4 is a factor of 4 (of course!) the equation (*) and the congruence posed will have solutions. We calculate

$$4 = 52 - 6 \times 8 = 52 - 6(60 - 52) = -6 \times 60 + 7 \times 52 ,$$

so (*) has a solution $x = 7$. Moreover, cancelling 4 from both sides of (*) gives the equation

$$13x - 15y = 1$$

in which the coefficients on the left hand side are coprime. So by the theorem on page 28 the general solution of (*) is $x = 7 - 15\lambda$, and the congruence has solution

$$x \equiv 7 \pmod{15} .$$

Comment. If you can spot right at the beginning that $\gcd(52, 60) = 4$, then the easiest way of all to solve (*) is to begin by cancelling 4 from both sides.

Note that in example 5 the modulus in the solution is different from that in the question. We'll investigate this further, but first a slightly different problem.

Problem. A notice at the bus stop says that “buses depart at x minutes past the hour, where $x \equiv 7 \pmod{15}$.” What does this mean in more customary language? How can the “customary language” be expressed as a congruence?

Answer. Buses leave at 7, 22, 37, 52 minutes past the hour, that is,

$$x \equiv 7, 22, 37 \text{ or } 52 \pmod{60} .$$

This congruence describes *exactly the same integers* x as the original congruence $x \equiv 7 \pmod{15}$.

Example 5, continued. Sometimes it is appropriate to write the solution to a congruence in terms of the same modulus as the original question. So we could write the solution of $52x \equiv 4 \pmod{60}$ as

$$x \equiv 7, 22, 37 \text{ or } 52 \pmod{60} .$$

Exercises.

- A notice at the central interstate coach terminal says that coaches leave every x hours after midnight, where $x \equiv 5 \pmod{8}$. Rewrite this in more sensible language, and as a congruence to a different modulus.
- Solve the congruence $9x \equiv -3 \pmod{24}$. Give your answer
 - as a congruence to the smallest possible modulus;
 - as a congruence modulo 24.

Notice from the previous five examples that a congruence may have no solution, one solution or more than one solution. The following result (which we shall not prove in lectures) shows when the different cases arise.

Theorem. *Number of solutions of congruences.* Consider the congruence $ax \equiv b \pmod{m}$.

- If a and m are relatively prime then the congruence has a unique solution modulo m .
- If $\gcd(a, m)$ is *not* a factor of b then the congruence has no solution.
- If $g = \gcd(a, m)$ is a factor of b then the congruence has a unique solution modulo m/g .
- If $g = \gcd(a, m)$ is a factor of b then the congruence has g different solutions modulo m .

Comments.

- If you would like to try proving these results, compare the theorems on pages 26 and 28.
- The third and fourth parts of the theorem really just give alternative ways of expressing the solution of the same congruence.
- The first part is actually a special case of the third and fourth! (Why?)

Exercise. Without actually solving, determine how many solutions the following congruences have. Give your answer in terms of the original modulus, and in terms of a smaller modulus if appropriate.

- $15x \equiv 18 \pmod{21}$
- $16x \equiv 19 \pmod{22}$
- $17x \equiv 20 \pmod{23}$

Answers.

With a bit more understanding of congruence properties we can sometimes solve congruences without using the Euclidean Algorithm and the Bézout property.

Theorem. *Cancellation in congruences.* Let a, b, c and m be integers, with $c \neq 0$.

- The congruences

$$ax \equiv b \pmod{m} \quad \text{and} \quad cax \equiv cb \pmod{cm}$$

have the same solutions.

- If $\gcd(c, m) = 1$, then the congruences

$$ax \equiv b \pmod{m} \quad \text{and} \quad cax \equiv cb \pmod{m}$$

have the same solutions.

Sketch of proof. Let $c \neq 0$. Then

$$m \mid ax - b \quad \text{if and only if} \quad cm \mid cax - cb ,$$

and this proves the first result. To prove the second we note firstly that it is always true that

$$\text{if } m \mid ax - b \quad \text{then} \quad m \mid cax - cb .$$

Moreover, $m \mid cax - cb$ can be rewritten as $m \mid c(ax - b)$; and if c and m are coprime this implies that $m \mid ax - b$.

These results may be interpreted as follows. We can cancel a factor from the left hand side and the right hand side of a congruence, *provided* that the factor is relatively prime to the modulus. We can also multiply both sides by a factor, *provided* that it is relatively prime to the modulus. We can cancel any factor from the left hand side and the right hand side of a congruence, regardless of coprimality, *provided* that we cancel it from the modulus as well.

Examples.

- We can solve example 5 very quickly as follows:

$$\begin{aligned}
 52x \equiv 4 \pmod{60} &\Leftrightarrow 13x \equiv 1 \pmod{15} \\
 &\Leftrightarrow -2x \equiv -14 \pmod{15} \\
 &\Leftrightarrow x \equiv 7 \pmod{15} .
 \end{aligned}$$

Exercise. Give a reason for each step in the above working.

- Another example: to solve $15x \equiv 2 \pmod{23}$ we have

$$\begin{aligned}
 15x \equiv 2 \pmod{23} &\Leftrightarrow 45x \equiv 6 \pmod{23} \\
 &\Leftrightarrow -x \equiv 6 \pmod{23} \\
 &\Leftrightarrow x \equiv -6 \pmod{23} .
 \end{aligned}$$

Exercise. Give reasons again.

- A different way to do the previous congruence begins

$$15x \equiv 2 \pmod{23} \Leftrightarrow 15x \equiv 25 \pmod{23} .$$

See if you can finish this.

Testing for primes. As mentioned above (page 16), it is possible to decide whether or not a number is prime without testing all potential divisors up to the square root – in fact, it can be done without testing any divisors at all!!

A very important result in number theory is the following.

Theorem. *Fermat's "Little" Theorem.* Let $p > 1$. If p is prime then for every integer a we have

$$a^p \equiv a \pmod{p} .$$

This can be restated as:

Corollary. Let $p > 1$. If there is any a such that

$$a^p \not\equiv a \pmod{p}$$

then p is composite.

Examples.

- We use this method to show that 201 is not prime. Take $a = 2$ and square it repeatedly, reducing modulo 201 each time, to get

$$\begin{aligned} 2^1 &\equiv 2 , & 2^2 &\equiv 4 , & 2^4 &\equiv 16 , & 2^8 &\equiv 55 , & 2^{16} &\equiv 10 \\ 2^{32} &\equiv 100 , & 2^{64} &\equiv 151 \equiv -50 , & 2^{128} &\equiv 88 \end{aligned}$$

modulo 201. Therefore

$$2^{201} = 2^{1+8+64+128} \equiv 8 \pmod{201} ,$$

and since $2^{201} \not\equiv 2 \pmod{201}$ we see that 201 is not prime.

- Let's try a harder example, $p = 341$. We can calculate powers as above, or more simply notice that

$$2^{10} = 1024 \equiv 1 \pmod{341}$$

and so

$$2^{341} = 2 \times 2^{340} \equiv 2 \times 1^{34} \equiv 2 \pmod{341}.$$

This leaves the question of whether or not 341 is prime unresolved. However, if it is prime then 3^{341} should equal 3 modulo 341. You can work out for yourself that

$$3^{341} \equiv 168 \not\equiv 3 \pmod{341},$$

and so 341 is not prime.

- The previous two examples are pretty pointless as it is easy to see that 201 is composite, and 341 is not very difficult either. However, consider $m = 2^{678} + 3^{789}$, which we looked at on page 15. Trying to completely factorise this is too hard even for **Maple** to do in a reasonable amount of time. Nevertheless we can use **Maple** to perform the calculations for Fermat's test. First enter

$$m := 2^{678} + 3^{789};$$

We now do *not* ask for $2^m \bmod m$; since then **Maple** will calculate the whole power 2^m before reducing modulo m , and this will take far too long. However the command

$$\text{Power}(2, m) \bmod m;$$

instructs **Maple** to use something similar to the “squaring” method from the first example, reducing at each step to keep the results small(ish); we almost immediately get an answer which is not 2, and this tells us that m is not prime. **Exercise.** Try it!

A drawback with this method of checking primality is that it only provides a *negative* test. That is, if an integer n fails the test it is definitely composite, but if it passes the test, *even for every possible value of a* , then it is “probably” but not certainly prime.

Consider, for example, $n = 561$. For every integer a we find that $a^{561} \equiv a \pmod{561}$. To confirm this we do not need to test all a from 0 to 560 but can reason as follows.

Observe that $561 = 3 \times 11 \times 17$. By Fermat’s Little Theorem we have

$$a^3 \equiv a \pmod{3}, \quad a^{11} \equiv a \pmod{11}, \quad a^{17} \equiv a \pmod{17}.$$

We can use these results to quickly evaluate a^{561} . We have

$$a^{561} \equiv (a^{11})^{51} \equiv a^{51} \equiv (a^{11})^4 a^7 \equiv a^4 a^7 \equiv a \pmod{11},$$

and we can do the same sort of thing modulo 17 (easier!) and modulo 3 (harder!) to get

$$a^{561} \equiv a \pmod{3}, \quad a^{561} \equiv a \pmod{11}, \quad a^{561} \equiv a \pmod{17}.$$

(**Exercise.** Fill in the details.) Since the moduli 3, 11, 17 are all relatively prime to each other, a^{561} is congruent to a modulo their product. That is, $a^{561} \equiv a \pmod{561}$ even though 561 is not prime.

A composite number which passes Fermat’s test for all a is called an **absolute pseudoprime** or a **Carmichael number**. Google these terms for more information, or study MATH2400 Finite Mathematics.

Public–key cryptography. An important method of encrypting communications relies on prime numbers and congruences. If I wish to provide a method for anyone to send me a secret message which will be meaningless if intercepted by an “enemy”, I begin by finding two large prime numbers p and q and choosing an “encryption exponent” m relatively prime to $(p - 1)(q - 1)$. I publish m and the product $r = pq$ for anyone to use. Typically, r will be 200 digits long or more.

To encrypt a message, the sender first turns it into a “plaintext” number b by some agreed method. For example,

HEREISMYCREDITCARDNUMBER

could be written as

$$b = 080518050919132503180504092003011804142113020518 .$$

She then sends the “ciphertext”

$$c \equiv b^m \pmod{r} ,$$

which with the aid of a computer is not difficult to calculate. Now it is possible to prove that if n is a “decryption exponent” satisfying

$$mn \equiv 1 \pmod{(p - 1)(q - 1)} \tag{*}$$

then

$$b \equiv c^n \pmod{r} , \tag{**}$$

so I can recover the original message by solving the congruence (*) and computing the power (**).

Security. To solve $(*)$ and find n is easy for me as I have set up the code and I know p and q . But anyone wanting to break into my code would first have to find p and q by factorising r , and although this is simple in principle, it is “computationally infeasible” to do it in any reasonable amount of time. (Remember m from page 15!) Here is a typical calculation using Maple.

```
> p:=nextprime(2^333+1081);
           p := 17498...31689                [101 digits]
> q:=nextprime(3^222+1081);
           q := 83352...56961                [106 digits]
> r:=p*q;
           r := 14585...37129                [207 digits]
> m:=5^99+2^44+2;
           m := 15777...72543                [70 digits]
> igcd(m,(p-1)*(q-1));
           1
> b:=080518050919132503180504092003011804142113020518:
> c:=Power(b,m) mod r;
           c := 81243...17980                [206 digits]
> msolve(m*n=1,(p-1)*(q-1)); assign(%);
           { n = 35059...35487 }              [206 digits]
> Power(c,n) mod r;
           80518050919132503180504092003011804142113020518
> ifactor(r);
```


RELATIONS

Mathematics studies many kinds of things. Probably the first kind of “mathematical object” that we all think of is numbers, but of course there are many others. Early in school you learned about lines, triangles, quadrilaterals, circles and other geometrical figures; later on you began to study functions. In MATH1131 you were introduced to vectors and matrices; in MATH1081 and MATH1231 you have met (or will meet) sets, graphs, vector spaces, linear transformations and eigenvalues.

However, it’s not much use having all these concepts unless we can say something about them! We can think of a **relation** as being a word or expression which enables us to say something about two numbers, lines, sets or other mathematical objects.

Probably the most fundamental relation in any area of mathematics is *equality*. If we say “ x equals y ” we are making a statement about x and y ; “equals” is the word which is expressing a fact about x and y , and therefore is an example of a relation.

If you are familiar with grammatical terminology, you can think of a relation as a “mathematical verb”.

The best way to become comfortable with the concept of a relation is to look at lots of examples. Try to write down as many relations as you can think of in various areas of mathematics: a few are given to start you off.

- **Basic arithmetic, algebra and number theory**

$$= \quad <$$

- **Geometry**

$$\parallel$$

- **Set theory**

$$= \quad \subseteq$$

To formally define a relation between two “objects” we need to specify what kind of objects we are talking about, and which objects are related to which. We can do this simply by listing related numbers as an ordered pair: for example, we shall record the statement “ $2 < 3$ ” by writing down the pair $(2, 3)$. Note that it has to be an *ordered* pair: the pair $(3, 2)$ would correspond to “ $3 < 2$ ”, which is a different statement. These ideas prompt the following definition.

Definition. A **relation on a set** A is a set of ordered pairs of elements from A , that is, a subset of $A \times A$.

Notation. Since, formally speaking, relations are sets, we shall often denote them by capital letters. Many relations, including (most of?) those on page 54, have their own special symbols; we shall also use some symbols such as $\sim$ and $\preceq$ for “generic” relations.

Examples.

- The relation “less than or equal to” on $\mathbb{N}$ can be written as the set of ordered pairs

$$\begin{aligned} L &= \{ (0, 0), (0, 1), (0, 2), \dots, (1, 1), (1, 2), (1, 3), \dots \} \\ &= \{ (a, b) \in \mathbb{N}^2 \mid \end{aligned} \} .$$

- We can define a relation R on $A = \{ 2, 3, 4, 5, 6 \}$ by

$$\begin{aligned} R &= \{ (a, b) \in A^2 \mid a \text{ is a factor of } b + 2 \} \\ &= \{ (2, 2), (2, 4), (2, 6), (3, 4), \\ &\quad (4, 2), (4, 6), (5, 3), (6, 4) \} . \end{aligned}$$

To specify that two particular elements are related in this way we can write, for example,

$$(5, 3) \in R \quad \text{or} \quad 5 R 3 .$$

There are broader concepts of relation which we mention, but will not deal with in this course. First, it is not necessary that the two related objects be “of the same type”. A natural example of this occurs when we say that a point is on a line – “is on” is a relation from a set of points to a set of lines.

Definition. A **relation from a set A to a set B** is a set of ordered pairs (a, b) where $a \in A$ and $b \in B$, that is, a subset of $A \times B$.

There are a few important relations which involve three or more objects.

Definition. A **ternary relation** between sets A, B and C is a subset of $A \times B \times C$, and a **quaternary relation** between sets A, B, C and D is a subset of $A \times B \times C \times D$. In general, an **n -ary relation** is a subset of $A_1 \times A_2 \times \cdots \times A_n$ for certain sets $A_1, A_2, \dots, A_n$.

Note. The “basic” type of relation defined previously will be referred to as a *binary relation* if we need to distinguish it from an n -ary relation.

Examples.

- We can regard the congruence $a \equiv b \pmod{m}$ as defining a ternary relation among the integers a, b and m . Alternatively, we could think of m as being a fixed number; the congruence would then express a *binary* relation between a and b .
- We use a ternary relation whenever we say that three points are...
- A well-known quaternary relation expresses the fact that four points are...

Relations can be specified in various ways. As we have seen, a relation is (formally) a set of ordered pairs, and can be defined by *listing* the pairs in the set, or by *describing* the pairs in the set.

There are two other useful ways of depicting a relation on a *finite* set: matrices and “arrow diagrams”. To construct the **matrix** M of a relation on a set A with n elements we first choose a specific order for these elements, say $A = \{a_1, a_2, \dots, a_n\}$. Then M is the $n \times n$ matrix in which the row i , column j element is

$$m_{i,j} = \begin{cases} 1 & \text{if } a_i \text{ is related to } a_j \\ 0 & \text{if } a_i \text{ is not related to } a_j. \end{cases}$$

The **arrow diagram** of the relation consists of a point for each element of A , with an arrow drawn from a_i to a_j if and only if a_i is related to a_j .

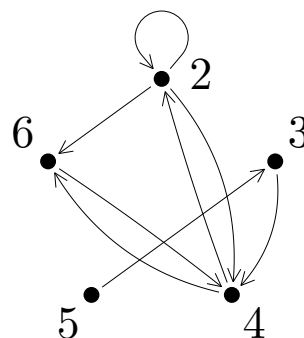
Examples.

- On page 55 we saw the relation R on $A = \{2, 3, 4, 5, 6\}$, defined by

$$\begin{aligned} R &= \{ (a, b) \in A^2 \mid a \text{ is a factor of } b + 2 \} \\ &= \{ (2, 2), (2, 4), (2, 6), (3, 4), \\ &\quad (4, 2), (4, 6), (5, 3), (6, 4) \} . \end{aligned}$$

If we list elements of A in the order shown, R has the matrix M below left. The arrow diagram for R is given below right.

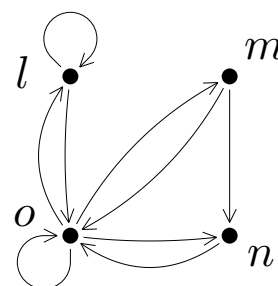
$$M = \begin{pmatrix} 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{pmatrix}$$



- Let $A = \{l, m, n, o\}$ and define a relation R on A as follows: $\alpha R \beta$ if and only if there is an English word beginning with the letters $\alpha\beta$. Then R can be written as a set of ordered pairs, as a matrix (taking the elements of A in alphabetical order) and as an arrow diagram.

$$R = \{ (l, l), (l, o), (m, n), (m, o), (n, o), (o, l), (o, m), (o, n), (o, o) \}$$

$$M = \begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{pmatrix}$$



Comments.

- Arrow diagrams are “officially” known as *directed graphs*, but you are not supposed to have heard about these yet!
- Since the elements of a set can (usually) be listed in various different orders, there is more than one possible matrix of a given relation.

Exercise. Write down the matrix of R if we take the elements of A in the order o, m, l, n .

There are various terms describing characteristics which are possessed by certain relations (and not by others). Four of these are especially important.

Definition. A relation R on a set A is said to be **reflexive** if for every $a \in A$ it is true that $a R a$. That is, if every element of A is related to itself.

Examples.

- The following relations are reflexive,

$=$

while the following are not:

$<$

- Is the relation R on page 59 reflexive?

Exercise. How can you pick a reflexive relation by looking at its matrix? By looking at its arrow diagram?

Definition. A relation R on a set A is said to be **symmetric** when for all a, b in A , if $a R b$ then $b R a$. That is, whenever one element is related to another, the second is also related to the first.

Examples.

- Some symmetric relations...

=

...and some not:

≤

- Two non-mathematical examples: on the set of all people, “...is a cousin of...” is a symmetric relation, but “...is a brother of...” is not.
- How about the relation from page 59?

Exercise. What does the symmetry property mean in a matrix? In a diagram?

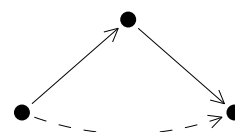
Definition. Let R be a relation on a set A . We say that R is **transitive** when for all $a, b, c \in A$, if $a R b$ and $b R c$ then $a R c$. In other words, if one element is related to a second and the second to a third, then the first element is related to the third.

Examples. Many well-known relations are transitive,

=

a few are not:

Diagrams and matrices. An arrow diagram represents a transitive relation if it never contains “two sides of a triangle” with the third side missing.



To determine whether the relation represented by a matrix M is transitive, calculate M^2 and look at the places where non-zero entries occur. If M has a 1 in *all* of these places then the relation is transitive; if M has a 0 in one or more of the places then it is not. For example, given the matrix

$$M = \begin{pmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$

of a relation on $\{1, 2, 3, 4\}$, we can calculate

$$M^2 = \begin{pmatrix} 2 & 0 & 1 & 3 \\ 2 & 1 & 2 & 2 \\ 1 & 0 & 0 & 1 \\ 2 & 0 & 1 & 2 \end{pmatrix}.$$

Examining these matrices, we see that in row 2, column 4 of M^2 there is a 2, while in the corresponding place of M there is a 0. So the relation is not transitive; in particular, we find that 2 is related to 1 and 1 is related to 4, but 2 is not related to 4.

Exercises.

- There are three further violations of transitivity in this example: find them. Also, confirm all four by drawing an arrow diagram for the relation.
- Can you explain why matrix multiplication is relevant to this problem?

Definition. Let R be a relation on a set A . Then R is said to be **antisymmetric** when for all $a, b \in A$, if $a R b$ and $b R a$ then $a = b$. That is, two *different* elements of A may be related by R one way around, or the other, or neither, but never both.

Warning! “Antisymmetric” does not mean “not symmetric”!

Examples. A few examples of antisymmetric relations...

=

...and some non-examples:

Notes.

- Since equality is both symmetric and antisymmetric, it is clear that “antisymmetric” is not the opposite of “symmetric”.
- The statement that $<$ is antisymmetric is

$$\text{if } a < b \text{ and } b < a \text{ then } a = b .$$

If you want to show that this is false, you have to find a and b for which

$$a < b \text{ and } b < a \text{ and } a \neq b .$$

This is impossible, and so the statement is true! If X is a statement which can *never* be true, then any statement of the form “if X then Y ” is automatically true, and is sometimes referred to as *vacuously true*.

Exercise. What properties of arrow diagrams and matrices characterise an antisymmetric relation?

Of the four properties reflexivity, symmetry, transitivity and antisymmetry we have just introduced, two combinations are particularly important.

Definition. A relation which is reflexive, symmetric and transitive is called an **equivalence relation**.

Examples.

Comment. Intuitively, an equivalence relation tells us when two things are “the same” – or, at least, the same in whatever respect interests us for a particular problem. If you can re-express a relation, using the word “same” in your answer, you can be reasonably sure it is an equivalence relation. Consider the following examples.

- Two numbers are equal if they are “the same” – equality can be regarded as the fundamental example of an equivalence relation.
- Two lines are parallel if they have the same direction.
- Two sets are equal if...
- Two triangles are similar if...
- Two triangles are congruent if...
- Two geometric figures are similar if...
- Two geometric figures are congruent if...
- Integers a and b are congruent modulo m if...

Example. A relation $\sim$ on $\mathbb{R}$, the set of real numbers, is defined by

$$x \sim y \quad \text{if and only if} \quad \cos x = \cos y .$$

Show that $\sim$ is an equivalence relation.

Proof. First, let $x \in \mathbb{R}$. Obviously $\cos x = \cos x$, and so $x \sim x$. Therefore $\sim$ is **reflexive**.

Next, let $x, y \in \mathbb{R}$ and assume that $x \sim y$. By definition this means that $\cos x = \cos y$. It follows immediately that $\cos y = \cos x$, that is, $y \sim x$. Hence $\sim$ is **symmetric**.

Finally, let $x, y, z \in \mathbb{R}$; suppose that $x \sim y$ and $y \sim z$. Then

$$\cos x = \cos y \quad \text{and} \quad \cos y = \cos z ;$$

therefore $\cos x = \cos z$, and so $x \sim z$. This proves that $\sim$ is **transitive**.

Since $\sim$ is reflexive, symmetric and transitive, it is an equivalence relation.

Comments.

- On an intuitive level we could argue that $\sim$ is an equivalence relation because x and y are related if... (exercise!). However, it is also important to be able to prove this formally.
- Essentially, $\sim$ is an equivalence relation because equality is an equivalence relation.
- **Note** that the algebra in this proof is absolutely trivial! The important thing is to make sure that you know what the words mean, and that you understand the logic of the proof.

Consider the relation on $\mathbb{Z}$ of congruence modulo 5, which is an equivalence relation. (**Exercise:** carefully check this!) We have

$$\dots \equiv -6 \equiv -1 \equiv 4 \equiv 9 \equiv 14 \equiv \dots \pmod{5} ;$$

in slightly inaccurate but convenient language we can say that $-6, -1, 4, 9, 14$ and so on are all “the same” modulo 5. Therefore, instead of regarding them as infinitely many different numbers we shall think of them as one *set* of numbers. As the “simplest” of these numbers is 4 we shall denote the set by $[4]$. That is,

$$\begin{aligned} [4] &= \{ x \in \mathbb{Z} \mid x \equiv 4 \pmod{5} \} \\ &= \{ \dots, -6, -1, 4, 9, 14, \dots \} . \end{aligned}$$

Looking at different collections of congruent numbers from the same point of view we can define

$$\begin{aligned} [0] &= \{ x \in \mathbb{Z} \mid x \equiv 0 \pmod{5} \} = \{ \dots, -5, 0, 5, 10, \dots \} \\ [1] &= \{ x \in \mathbb{Z} \mid x \equiv 1 \pmod{5} \} = \{ \dots, -4, 1, 6, 11, \dots \} \\ [2] &= \{ x \in \mathbb{Z} \mid x \equiv 2 \pmod{5} \} = \{ \dots, -3, 2, 7, 12, \dots \} \\ [3] &= \{ x \in \mathbb{Z} \mid x \equiv 3 \pmod{5} \} = \{ \dots, -2, 3, 8, 13, \dots \} . \end{aligned}$$

In fact, we could do all this for any integer m , but we would always find that the result was one of the five sets we have obtained already. For example,

$$\begin{aligned} [6] &= \{ x \in \mathbb{Z} \mid x \equiv 6 \pmod{5} \} \\ &= \{ \dots, -4, 1, 6, 11, 16, \dots \} \\ &= \{ x \in \mathbb{Z} \mid x \equiv 1 \pmod{5} \} \\ &= [1] . \end{aligned}$$

If desired, we could now completely cease to write about integers and congruence modulo 5, and instead put everything in terms of sets and set equality. We would have to specify what we mean by adding and multiplying sets, so say that by definition

$$[a] + [b] = [a + b] \quad \text{and} \quad [a][b] = [ab] .$$

Then instead of a calculation like

$$2 \times 3 + 4 \equiv 6 + 4 \equiv 1 + 4 \equiv 5 \equiv 0 \pmod{5}$$

we would write

$$[2][3] + [4] = [6] + [4] = [1] + [4] = [5] = [0] ,$$

essentially the same calculation in a different notation.

Observe that while it is not accurate to write $6 = 1$ (we should write $6 \equiv 1 \pmod{5}$ instead), it is absolutely correct to write $[6] = [1]$: here the symbol $=$ denotes equality of *sets*, not equality of numbers.

We introduce some terminology to describe what we have just been doing.

Definition. Let $\sim$ be an equivalence relation on a set A . For any a in A , the **equivalence class of a with respect to $\sim$** is the set

$$[a] = \{x \in A \mid x \sim a\} .$$

We have seen that an equivalence relation tells us when two objects are in some sense “the same”; an equivalence class, then, collects together all the objects which are “the same” so that we can regard them as a single “object”.

Examples.

- If $\sim$ is $\equiv (\text{mod } 5)$ then there are five equivalence classes, namely, the sets $[0], [1], [2], [3], [4]$ that we considered above.
- Consider the set A of all lines in a plane. The relation $\parallel$ on A is an equivalence relation, and the equivalence class of any line ℓ is

$$[\ell] = \{k \in A : k \parallel \ell\} = \{\text{all lines parallel to } \ell\} .$$

- Likewise, if B is the set of all triangles in a plane and we consider the relation of similarity, the equivalence class of any particular triangle is the set of all triangles similar to that one. For example, the equivalence class of an equilateral triangle is the set of *all* equilateral triangles, and the equivalence class of a right-angled isosceles triangle is the set of all right-angled isosceles triangles. **Exercise.** True or false? – the equivalence class of a specific isosceles triangle is the set of all isosceles triangles.

Exercise. Consider the relation on $\mathbb{R}$ from page 67,

$$x \sim y \quad \text{if and only if} \quad \cos x = \cos y .$$

Find $[0]$ and $[1]$; then find a general formula for the equivalence class $[a]$.

Solution.

Examples.

- On $A = \{2, 3, 4, \dots, 20\}$ we define $x \sim y$ if and only if x and y have the same prime factors, ignoring repetitions. The equivalence classes with respect to this relation are

$$[2] = \{2, 4, 8, 16\}$$

$$[3] = \{3, 9\}$$

$$[5] = \{5\}$$

$$[6] = \{6, 12, 18\}$$

$$[7] = \{7\}$$

$$[10] = \{10, 20\}$$

$$[11] = \{11\}$$

$$[13] = \{13\}$$

$$[14] = \{14\}$$

$$[15] = \{15\}$$

$$[17] = \{17\}$$

$$[19] = \{19\} .$$

- Let $S = \{\text{students currently enrolled in 1081}\}$, and say that $x \sim y$ if and only if x and y have surnames with the same initial. How many equivalence classes does this relation have?

If you look carefully at some of the previous examples, you may notice two important facts about equivalence classes: they never overlap, and between them they contain all of the “objects” under discussion.

Theorem. Let $\sim$ be an equivalence relation on a set A . Then the equivalence classes with respect to $\sim$ *partition* A : that is,

- every element of A is in some equivalence class;
- no element of A is in two different classes.

Or to put it another way: the equivalence classes are disjoint, and their union equals A .

Proof. It is easy to show that every element of A is in some equivalence class. If $a \in A$ then $a \sim a$ because $\sim$ is reflexive; therefore $a \in [a]$.

Now suppose that an element a is in two classes, say $a \in [b]$ and $a \in [c]$. We shall show that $[b] = [c]$: that is, a cannot be in two *different* classes. So, let x be an element of $[b]$. We have

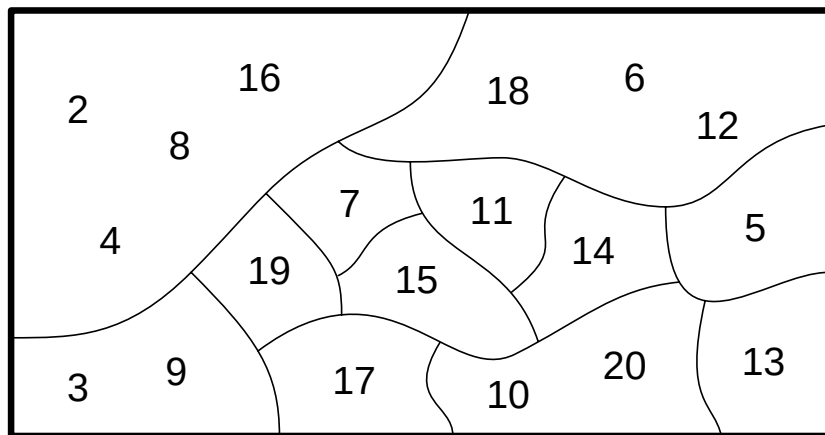
$$x \sim b \quad \text{and} \quad a \sim b \quad \text{and} \quad a \sim c ;$$

since $\sim$ is symmetric

$$x \sim b \quad \text{and} \quad b \sim a \quad \text{and} \quad a \sim c ;$$

and using (twice) the fact that $\sim$ is transitive, we have $x \sim c$. This means that $x \in [c]$, so we have shown that $[b] \subseteq [c]$. We can prove similarly that $[c] \subseteq [b]$; therefore $[b] = [c]$, and this is what we wanted to prove.

Example. We can illustrate the equivalence classes of the first relation on page 72 as follows.



Definition. A relation which is reflexive, antisymmetric and transitive is called a **partial order**.

Examples.

Comments.

- Roughly, a partial order tells us which of two elements comes first (in some sense). For example, let $a, b \in \mathbb{R}$. Then
 - $a \leq b$ means that a comes before b , if we list elements in increasing order;
 - $a \geq b$ means that a comes before b , if we list elements in decreasing order.
- A partial order may have an additional property,

$$\text{for all } a, b \in A, \quad \text{either } a \preceq b \quad \text{or} \quad b \preceq a . \quad (*)$$

For example,

- relations with this property include...
- relations without this property include...

If $\preceq$ is a partial order for which $(*)$ is true we refer to it as a *total order* (because any two elements can be ordered) or a *linear order* (because the elements can be illustrated as lying in order along a line).

- The term “poset” is sometimes used for a set on which a partial order has been defined.

Example. On $\mathbb{C}$, the set of complex numbers, we define

$$w \preceq z \quad \text{if and only if} \quad \operatorname{Re} w \leq \operatorname{Re} z \text{ and } \operatorname{Im} w \leq \operatorname{Im} z .$$

Prove that $\preceq$ is a partial order.

Proof. Let $z \in \mathbb{C}$. Obviously $\operatorname{Re} z = \operatorname{Re} z$ and $\operatorname{Im} z = \operatorname{Im} z$. Therefore $\operatorname{Re} z \leq \operatorname{Re} z$ and $\operatorname{Im} z \leq \operatorname{Im} z$. So $z \preceq z$, and we have proved that $\preceq$ is **reflexive**.

Next, let $w, z \in \mathbb{C}$; suppose that $w \preceq z$ and $z \preceq w$. Then

$$\begin{aligned} \operatorname{Re} w &\leq \operatorname{Re} z & \text{and} & & \operatorname{Im} w &\leq \operatorname{Im} z , \\ \operatorname{Re} z &\leq \operatorname{Re} w & \text{and} & & \operatorname{Im} z &\leq \operatorname{Im} w . \end{aligned}$$

It follows that $\operatorname{Re} w = \operatorname{Re} z$ and $\operatorname{Im} w = \operatorname{Im} z$; therefore $w = z$. Hence, $\preceq$ is **antisymmetric**.

Finally, let $z_1 \preceq z_2$ and $z_2 \preceq z_3$. This means that

$$\begin{aligned} \operatorname{Re} z_1 &\leq \operatorname{Re} z_2 & \text{and} & & \operatorname{Im} z_1 &\leq \operatorname{Im} z_2 , \\ \operatorname{Re} z_2 &\leq \operatorname{Re} z_3 & \text{and} & & \operatorname{Im} z_2 &\leq \operatorname{Im} z_3 . \end{aligned}$$

Hence $\operatorname{Re} z_1 \leq \operatorname{Re} z_3$ and $\operatorname{Im} z_1 \leq \operatorname{Im} z_3$, that is, $z_1 \preceq z_3$. Therefore $\preceq$ is **transitive**.

Since $\preceq$ is reflexive, antisymmetric and transitive, it is a partial order.

Comments.

- As we saw before, if you know what the words mean, and take care with the logic and setting out of your answer, you should find you do not have much trouble with this kind of problem.
- Your teachers in school (and MATH1131/MATH1141) have probably told you that complex numbers cannot be ordered!! So what is going on here??

Example. On $\mathbb{R}$ we define $x \preceq y$ if and only if

$$y = x + k \text{ for some non-negative integer } k.$$

Show that $\preceq$ is a partial order.

Proof. Let $x \in \mathbb{R}$; then $x = x + 0$, and 0 is a non-negative integer; so $x \preceq x$. Therefore $\preceq$ is reflexive.

Let $x, y \in \mathbb{R}$; suppose that $x \preceq y$ and $y \preceq x$. Then

$$y = x + k \quad \text{and} \quad x = y + l$$

for some non-negative integers k, l . Adding these equations,

$$y + x = x + k + y + l$$

and so $k + l = 0$. Since k, l cannot be negative, the only possibility is that $k = 0$ and $l = 0$. Hence $x = y$, and this shows that $\preceq$ is antisymmetric.

Let $x, y, z \in \mathbb{R}$; suppose that $x \preceq y$ and $y \preceq z$. Then

$$y = x + k \quad \text{and} \quad z = y + l$$

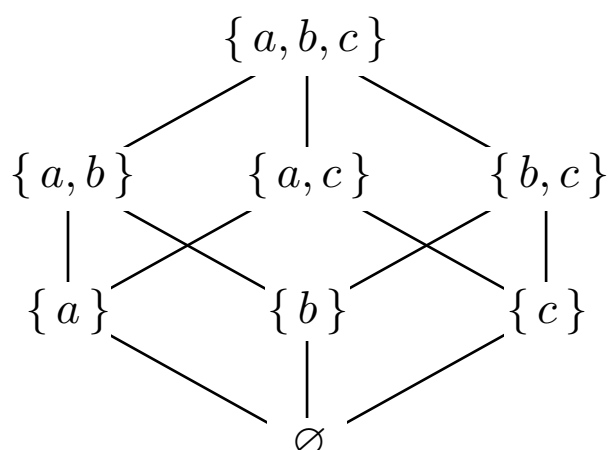
for some non-negative integers k, l . Eliminating y , we have

$$z = (x + k) + l = x + (k + l) ;$$

since $k + l$ is a non-negative integer, $x \preceq z$. Therefore $\preceq$ is transitive.

Since $\preceq$ is reflexive, antisymmetric and transitive, it is a partial order.

We can usefully illustrate a partial order on a *finite* set by means of a **Hasse diagram**. For example, consider the relation $\subseteq$ on $\mathcal{P}(\{a, b, c\})$, the set of all subsets of $\{a, b, c\}$. This partial order has the following Hasse diagram.



In this Hasse diagram, if two sets S and T are related, that is, $S \subseteq T$, then we place S further down the page than T and draw a line between them, **except** that we simplify the diagram in two respects.

- We have always $S \subseteq S$, because a partial order is reflexive. So we allow any “loops” in the diagram to be understood and do not actually draw them.
- We do not draw any lines which can be deduced by transitivity. For example, it is true that $\{a\} \subseteq \{a, b, c\}$, but we have not drawn a line between these two sets. Why not? – because we can see from the diagram that

$$\{a\} \subseteq \{a, b\} \quad \text{and} \quad \{a, b\} \subseteq \{a, b, c\} ,$$

and since a partial order is always transitive, this tells us that $\{a\} \subseteq \{a, b, c\}$.

Comments.

- We can formally define the Hasse diagram of a partial order $\preceq$ on a finite set A as follows. First, write $a \prec b$ to mean that $a \preceq b$ but $a \neq b$. Then the Hasse diagram consists of
 - a point for each element of A , with a occurring “further down the page” than b if $a \prec b$;
 - a line from a to b if and only if $a \prec b$ and there is *no* element c such that $a \prec c \prec b$.
- Given the Hasse diagram of a partial order $\preceq$, we can tell that $a \preceq b$ if there is a sequence of lines *always leading upwards* from a to b .
- What does the diagram on the previous page look like if you remove all the sets and just concentrate on the lines? Can you explain this?

Exercise. Draw the Hasse diagram for the partial order on

$$S = \{ (1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2), (4, 1), (4, 2), (4, 3) \}$$

defined by

$$(a, b) \preceq (c, d) \quad \text{if and only if} \quad a \leq c \text{ and } b \leq d .$$

Answer.

Definition. Let $\preceq$ be a partial order on a set A . An element x of A is called

- **greatest** if every element is related to it, that is, $a \preceq x$ for all $a \in A$;
- **least** if it is related to every element, that is, $x \preceq a$ for all $a \in A$;
- **maximal** if it is related to *no* element except itself, that is, $x \preceq a$ *only* when $a = x$;
- **minimal** if no element except itself is related to it, that is, $a \preceq x$ *only* when $a = x$.

Examples. In the partially ordered set on the previous page,

- the greatest element(s) is/are...
- the least element(s) is/are...
- the maximal element(s) is/are...
- the minimal element(s) is/are...

Note. Make sure you understand the difference between *greatest* and *maximal* elements, and between *least* and *minimal*.

Theorem. *Properties of greatest, least, maximal and minimal elements in a partial order.* In a partial order on a set A ,

- any greatest element is maximal, and any least element is minimal;
- there is only one greatest element (if any), and only one least element (if any).

Also, suppose that A is a finite set. Then

- if there is a unique maximal element then it is the greatest element;
- if there is a unique minimal element then it is the least element.

Exercises.

- Prove the above properties.
- Find an example of a partial order on an *infinite* set for which the last two properties are not true. **Suggestion:** don't look for any simple "rule", just draw some kind of Hasse diagram.

Definition. Let $\preceq$ be a partial order on a set A , and let $a, b \in A$. Then

- any $x \in A$ such that $x \preceq a$ and $x \preceq b$ is called a **lower bound** of a and b ;
- any $x \in A$ such that $a \preceq x$ and $b \preceq x$ is called an **upper bound** of a and b .

The **greatest lower bound** of a and b , if there is one, and the **least upper bound**, if there is one, are denoted by

$$\text{glb}(a, b) \quad \text{and} \quad \text{lub}(a, b) .$$

Examples. In the partial order with Hasse diagram on page 80,

- the upper bounds of $(1, 2)$ and $(3, 1)$ are...
- the least upper bound of $(1, 2)$ and $(3, 1)$ is...
- the lower bounds of $(2, 3)$ and $(3, 2)$ are...
- the greatest lower bound of $(2, 3)$ and $(3, 2)$ is...

Exercise. Draw the Hasse diagram for the divisibility relation on the set

$$S = \{ \text{factors of } 72 \} = \{ 1, 2, 3, 4, 6, 8, 9, 12, 18, 24, 36, 72 \} .$$

Does this set have greatest, least, maximal, minimal elements? Show that any two elements have a greatest lower bound and a least upper bound. What are $\text{glb}(a, b)$ and $\text{lub}(a, b)$?

Remove the elements 6, 24 and 72 from S . How does this affect your answers? Can you find two numbers which have no greatest lower bound? No least upper bound?

Answers.

